

Scaling behaviors in simulated collagen fibrils

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Abstract

Collagen fibrils provide the primary source of tensile strength in mammalian connective tissues. Understanding how assembly kinetics dictate fibril architecture—and how that topology governs mechanical failure—is central to soft matter physics and bio-inspired material design. Here, we model collagen fibril formation using a modified diffusion-limited aggregation (DLA) framework where rod-like molecules undergo T_s surface-diffusion steps post-attachment, progressively compacting the structure. Coupling this morphological model with a probabilistic fracture framework reveals that T_s drives a structural crossover from porous fractal cross-sections to nearly space-filling architectures, with the fractal dimension D_f rising monotonically from 1.708 ± 0.005 ($T_s = 2$) to 1.963 ± 0.001 ($T_s = 8192$). This compaction increases the mean rupture force F_r by an order of magnitude, with 83% of the strength gain already realized at $T_s = 128$. Mechanistically, failure is governed by two local structural variables set by D_f : the mean load-bearing cross-sectional count $\langle N \rangle$ and the molecular coordination number $\langle K \rangle$. In open fibrils ($D_f \approx 1.71$), severe cross-sectional heterogeneity induces localized stress concentrations and rapid $\langle K \rangle$ loss, sustaining extensive failure cascades ($\gamma = 2.00 \pm 0.013$). In compact fibrils ($D_f \rightarrow 2.0$), uniform packing preserves $> 97\%$ of $\langle K_0 \rangle$ until failure, suppressing large cascades and steepening the power-law exponent to $\gamma = 2.55 \pm 0.008$. Across all architectures, pre-terminal damage remains bounded while final failure occurs via a single catastrophic cascade. These findings establish D_f as an architectural bridge linking assembly kinetics to failure statistics, offering a mechanistic principle for designing resilient biological and synthetic fibers.

I. INTRODUCTION

Type I collagen is the principal structural protein of the mammalian extracellular matrix (ECM), providing essential mechanical integrity to tissues such as skin, tendon, bone, cartilage, and the cornea [1–7]. These hierarchical assemblies are critical for elasticity, tensile strength, and resilience against physical stress [1, 8]. Individual collagen molecules, characterized by a high aspect ratio ($\approx 300 \text{ nm} \times 1.5 \text{ nm}$) [1, 9], self-assemble into fibrils that can reach lengths of $500 \text{ }\mu\text{m}$ and diameters of 500 nm [10–12]. These fibrils further aggre-

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gate into fibers (1–20 μm diameter), forming a multiscale network capable of storing and transmitting mechanical energy [7, 13].

Characterizing the mechanical behavior of collagen fibrils, particularly at the nanoscale, presents significant challenges [14]. Experimental techniques such as atomic force microscopy (AFM) and micro-mechanical actuation have successfully probed the tensile properties of isolated fibrils. Van der Rijt et al. used AFM to measure the mechanical properties of tendon fibrils, evaluating their tensile behavior until failure [15], while Svensson et al. [16] developed a piezoelectric actuator-based technique to measure stress and strain during uniaxial stretching. However, resolving the collective molecular dynamics that govern failure remains difficult due to experimental resolution limits, particularly when characterizing small-scale forces and intermolecular interactions [17]. Consequently, computational modeling has become indispensable for exploring the relationship between intrafibrillar organization and macroscopic mechanical response under a wider range of conditions.

Several theoretical frameworks have been developed to address fibril mechanics. Buehler et al. introduced coarse-grained molecular dynamics to model large-scale deformation, while Depalle et al. investigated the role of mineralization in stabilizing fibril elasticity in bone [18–22]. Other stochastic approaches have examined how enzymatic degradation alters tissue mechanics [23] and how morphological features like waviness influence fiber modulus [24]. Despite these advances, the role fractal organization may play in the mechanical resilience and the statistical dynamics of rupture remains largely unexplored.

In this work, we investigate how microscopic internal parameters—specifically the cross-sectional element count and neighbor topology—and the macroscopic fractal dimension jointly govern the failure behavior of collagen fibrils. We employ a modified Diffusion-Limited Aggregation (DLA) model incorporating a surface diffusion process, where the number of diffusion steps T_s minimizes the exposed surface area and regulates lateral compaction [25]. By coupling this morphological model with a probabilistic fracture framework, we demonstrate that T_s dictates both local structural features and global fractal geometry, which together determine the mechanical strength of the fibril. Despite its inherently stochastic nature, the model effectively captures the critical role of structural compactness and reveals stochastic avalanche dynamics during progressive failure in a driven disordered fracture process [26, 27].

II. MODEL AND RESULTS

A. Fibril Formation Model

We modeled collagen fibril formation using a three-dimensional diffusion-limited aggregation (DLA) algorithm on a cubic lattice. Collagen molecules were represented as rigid rods (rectangular bars) with dimensions $L_x = 1$, $L_y = 18$, and $L_z = 1$ in lattice units (l.u.), with L_y defining the characteristic length scale of the model.

The aggregation begins by placing the first molecule at the lattice center. Subsequent molecules are released one at a time from random positions on a sphere of radius R centered at the origin. The radius R is updated to track the maximum extent of the growing aggregate along the y -axis. Each molecule then performs a random walk on the lattice until one of the following conditions is met:

1. The molecule attaches to the existing aggregate;
2. The molecule is discarded if its distance from the origin exceeds $2R$.

After each attachment, R is recalculated. For simplicity, molecular rotation during diffusion is not allowed. This cycle of release, diffusion, and attachment continues until the aggregate reaches the desired size, limited here to 3×10^4 molecules.

Diffusion is implemented as a three-dimensional random walk on the cubic lattice. In the x - z plane, molecules can move to first-neighbor sites (1 l.u.) or second-neighbor sites ($\sqrt{2}$ l.u., i.e., diagonal moves). Along the y -axis, movement is restricted to first-neighbor sites (1 l.u.). In real collagen fibrillogenesis, lateral aggregation occurs with a characteristic axial spacing of 67 nm [11], which constrains the positions where new molecules can be incorporated. In our model, molecular attachment is restricted to multiples of 4 l.u. along the y -axis [25].

The formation of real fibrils is entropy-driven, by the release of water bound to the molecular surface, which favors lateral association and minimizes the exposed surface area [25, 28]; electrostatic interactions modulate this association [29, 30]. To mimic this relaxation, we implemented a lateral surface-diffusion step after attachment: a newly incorporated molecule performs a random walk restricted to the x - z plane (no movement along the y -axis), using the same rules described above. The molecule is allowed up to T_s diffusion attempts to search

the local surface and is placed at the position that minimizes its exposed area; if multiple positions are equivalent, the first one encountered is retained [31]. Using this algorithm, we generated ensembles of fibrils for different values of T_s to examine how post-attachment mobility controls fibril compactness, structure, and subsequent mechanical response.

The parameter T_s specifies the number of lateral surface-diffusion trials per deposition event, as each molecule completes its surface relaxation before the next addition. Consequently, T_s represents the ratio between the timescales of deposition and surface hopping. This sequential regime aligns with estimates by Parkinson et al. [25], showing that at the critical concentration for in vitro fibril formation, only on the order of ten collagen molecules are close enough to collide with a given surface region per second. Molecules thus arrive in isolation, and conditions that accelerate incorporation relative to surface mobility correspond to a smaller effective T_s .

The generated fibrils exhibit complex, heterogeneous packing with internal voids whose prevalence depends strongly on T_s (Fig. 1), which represents the number of diffusion attempts a molecule undergoes to minimize the external surface area and fill internal gaps. For $T_s = 2$, the fibril displays an open architecture characterized by large, visible gaps between molecular clusters. For intermediate values of T_s , the structure undergoes a progressive compaction as molecules more effectively find optimal sites to fill internal voids. Consequently, for $T_s = 8192$, the fibril becomes substantially more compact and homogeneous, with drastically reduced void space.

Although fibril growth proceeds primarily along the longitudinal axis (y -direction), the influence of the surface diffusion parameter T_s is most pronounced in the radial organization of the structure (x - z plane), as illustrated in Fig. 2. At low T_s , the projected cross-sections are open and branched. As T_s increases, however, the packing becomes progressively more compact and regular, reaching a highly uniform configuration at $T_s = 8192$. Since a higher local packing density implies a greater number of intermolecular contacts, this morphological transition suggests that fibrils formed at larger T_s exhibit enhanced connectivity and mechanical stability [32].

To quantify how T_s modulates fibril structure, we first estimated the fractal dimension of the central segment, defined as the region extending from $y = -100$ to $y = 100$. For each T_s , we generated 50 fibrils and isolated this central region from each. Within each central segment, we extracted 11 cross-sections at intervals of 18 l.u. between $y = -90$ and $y = 90$,

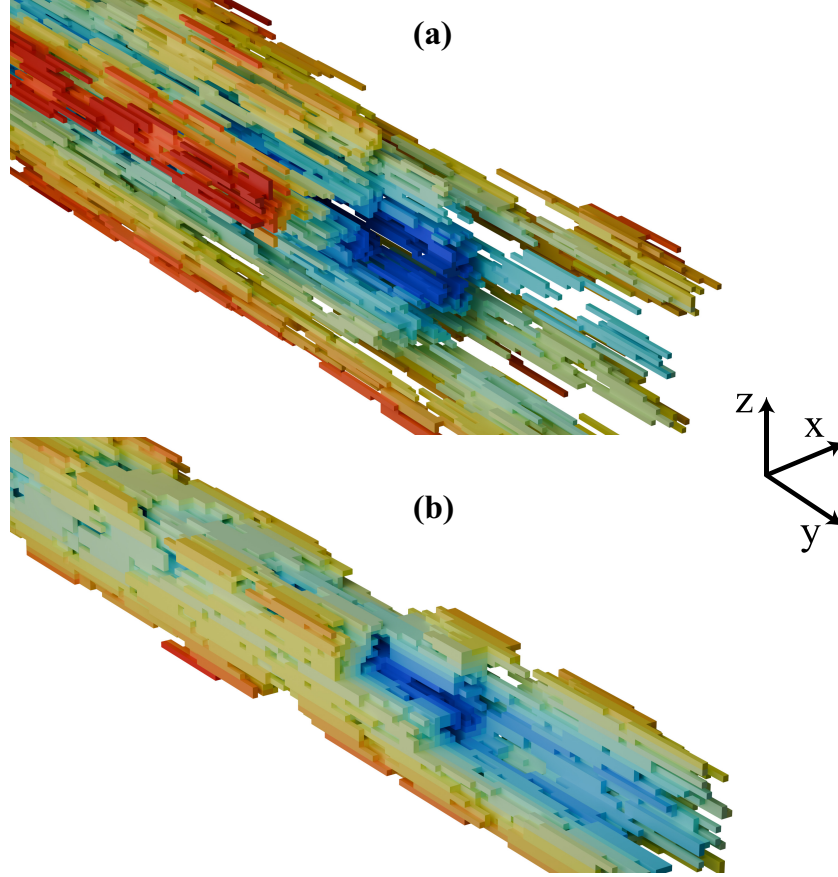


Figure 1. Representative fibril segments generated with the DLA algorithm containing 30,000 molecules. Panels (a) and (b) show typical fibrils obtained with $T_s = 2$ (short-term diffusion) and $T_s = 8192$ (long-term diffusion), respectively. Colors indicate the relative time of attachment, from early (blue) to late (red).

yielding a total of 550 cross-sections per T_s value. For each cross-section, we computed the center of mass and the maximum radius enclosing all particles (segments of molecules); $R_{\max}$ was subsequently defined as the largest of these 550 radii. The mass $m(R)$ was measured as the number of particles within a circular region of radius R centered at the cross-section's center of mass. For each value of R spanning from $R_{\min} = 5$ l.u. to $R_{\max}$, we computed the mean mass $\langle m(R) \rangle$ by averaging over all 550 cross-sections [33]. The fractal dimension D_f was then estimated from the mass-radius scaling relation [34, 35]:

$$\langle m(R) \rangle \sim R^{D_f}, \quad (1)$$

by performing a linear fit on a log-log plot of $\langle m(R) \rangle$ versus R .

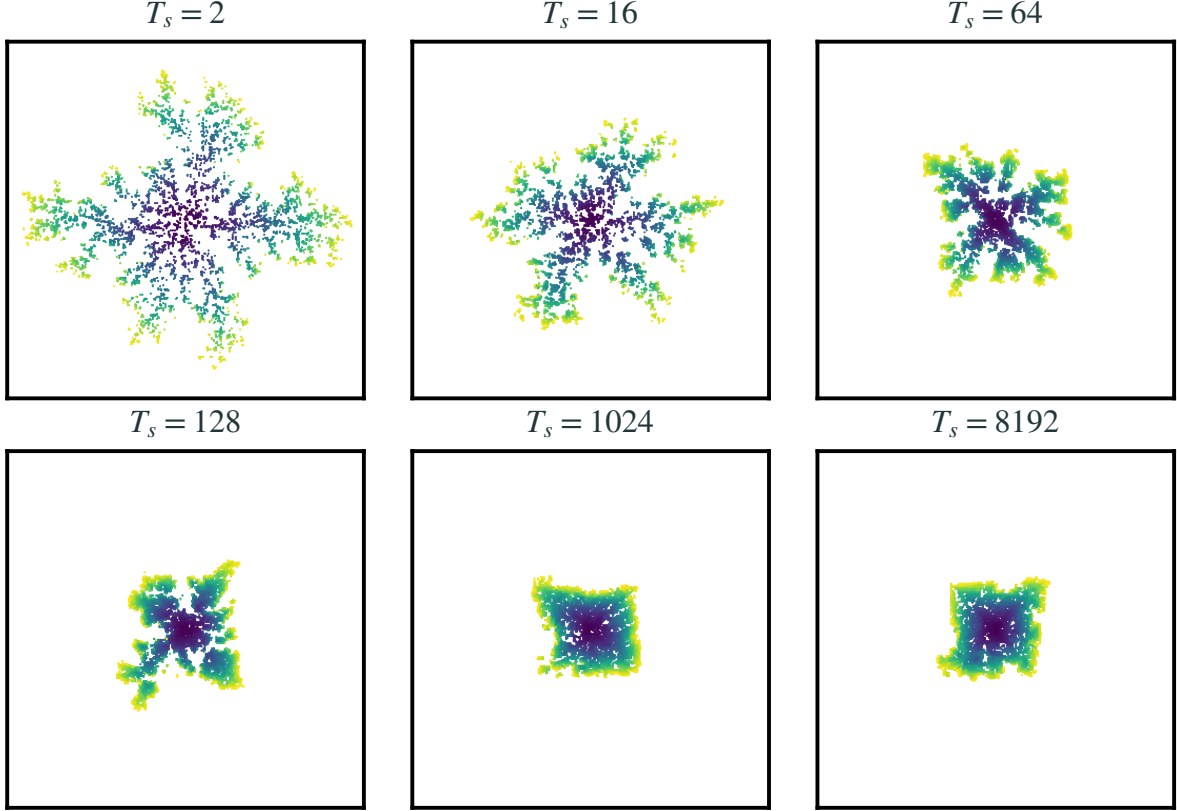


Figure 2. Representative projected central segments (x - z plane) for $T_s = 2, 16, 64, 128, 1024$, and 8192 . Increasing T_s illustrates a clear transition from a sparse, irregular morphology at low values to dense, radially symmetric packing at high values. The color gradient (blue to yellow) indicates the temporal sequence of molecular incorporation, from early-attached to recently added molecules.

As shown in Fig. 3, D_f depends strongly on T_s , providing a quantitative measure of the morphological transition observed in Fig. 2. For $T_s = 2$, we find that $D_f = 1.708 \pm 0.005$, close to the characteristic value of two-dimensional diffusion-limited aggregation ($D_f \approx 1.71$) [36]. As T_s increases, D_f rises and saturates at $D_f = 1.963 \pm 0.001$ for $T_s = 8192$, approaching the Euclidean limit of $D_f = 2.0$ expected for a compact two-dimensional disk. Thus, increasing post-attachment mobility drives a crossover from diffusion-limited, fractal packing to nearly space-filling cross-sections.

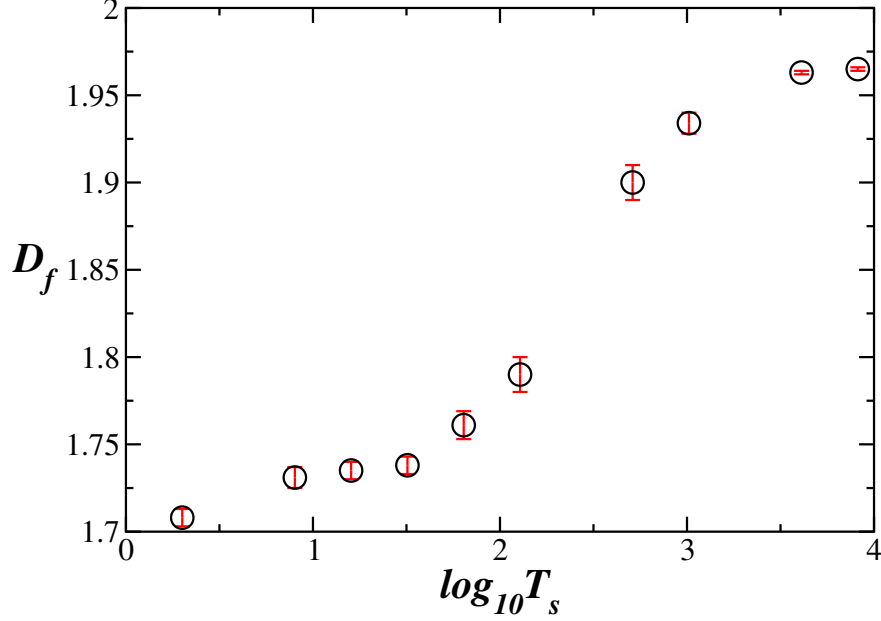


Figure 3. Average fractal dimension D_f as a function of the diffusion parameter T_s on a log-linear plot. Error bars are shown in red. The fractal dimension increases from $D_f = 1.708 \pm 0.005$ at $T_s = 2$ to $D_f = 1.963 \pm 0.001$ for $T_s = 8192$.

B. Mechanical Model for Fibril Rupture

To connect the T_s -dependent fibril architecture to mechanical failure, we modeled rupture using a progressive, probabilistic fracture framework commonly employed for disordered networks [37–39]. Because collagen molecules are represented as rigid rods that do not undergo continuous deformation prior to failure, we do not attempt to resolve deterministic crack propagation. Instead, intermolecular bonds fail stochastically under an applied axial force, and whole molecules detach from the load-bearing network with a stress-dependent probability. The procedure consists of: (i) isolating a trunk (force-carrying core) with dimensions $S_x = 17$, $S_y = 201$, and $S_z = 17$ l.u. [40]. Truncation inevitably creates disconnected dangling ends that do not contribute to force transmission [41]; we therefore (ii) identify the backbone (active skeleton) within the trunk prior to applying tensile loading [42].

Backbone extraction proceeds in two passes. In the first pass, molecules at $y = 1$ are marked as active, and we scan upward to $y = S_y$, progressively marking molecules in contact with active ones; unmarked molecules are then removed. In the second pass, molecules at

¹⁴⁴ $y = S_y$ are marked as active, and we scan downward to $y = 1$ using the same connectivity
¹⁴⁵ rule; only molecules marked active after this pass are retained, defining the final backbone
¹⁴⁶ [40] (Fig. 4).

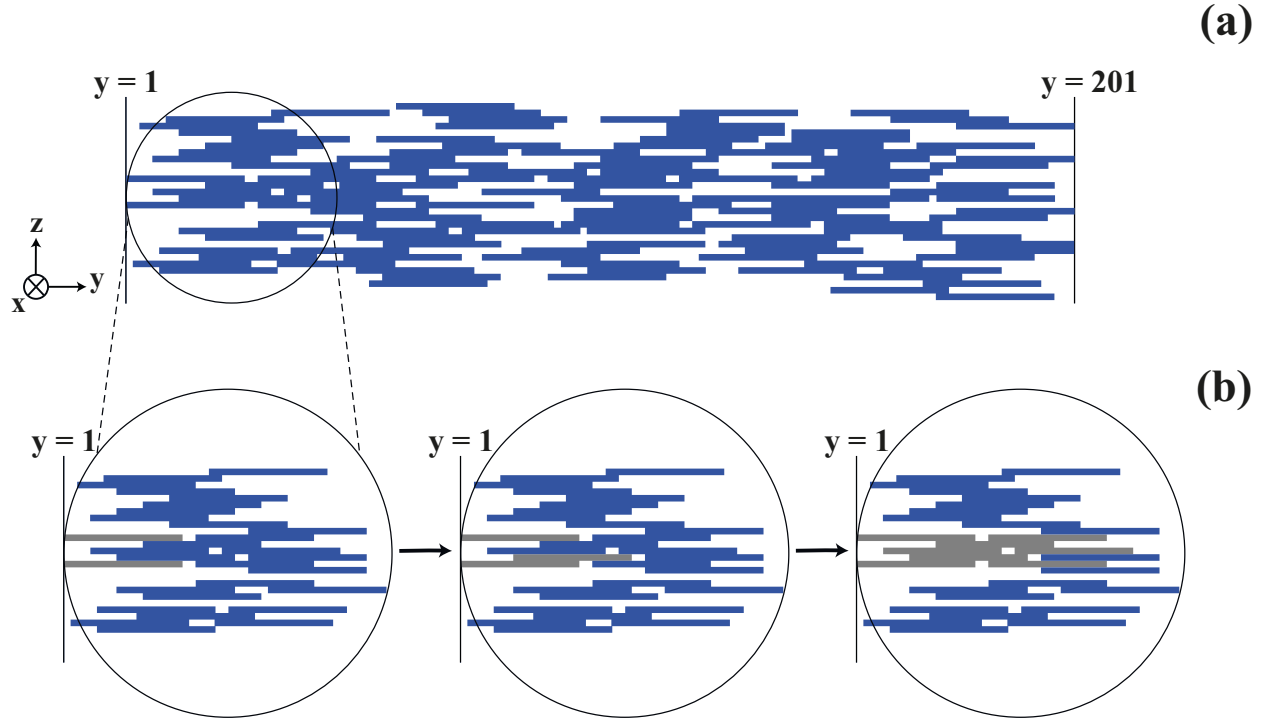


Figure 4. (a) Two-dimensional visualization of an isolated fibril trunk used to extract the backbone. The circle highlights the region where the identification procedure starts ($y = 1$). (b) Example of the backbone identification procedure from left to right: molecules marked as active are highlighted in gray as connectivity is propagated along the trunk, yielding the final backbone.

¹⁴⁷ With the backbone identified, we simulate the effects of axial tension along the fibril's
¹⁴⁸ axis (y) by progressively increasing a dimensionless force parameter F , where F regulates
¹⁴⁹ the removal of molecules in a probabilistic fashion. At each cross-section i , we define the
¹⁵⁰ effective load-bearing area as the number of backbone segments intersecting that plane,
¹⁵¹ $N(i)$. Each discrete segment is assigned a unit area on the lattice [25]; assuming uniform
¹⁵² load sharing, the local tensile stress at cross-section i is:

$$\sigma(i) = \frac{F}{N(i)}. \quad (2)$$

¹⁵³ We assume that local failure occurs by rupturing intermolecular bonds so that molecules
¹⁵⁴ detach from each other rather than break [40]. For a molecule spanning n cross-sections,

the mean stress is computed as the average of the local cross-sectional stresses:

$$\sigma_M = \frac{1}{n} \sum_{i=1}^n \sigma(i). \quad (3)$$

A molecule's resistance to detach is assumed to scale with the number of intermolecular contacts it forms within the backbone [43]. For each molecule, we count first-neighbor contacts for every segment in each intersected cross-section and sum these contributions to obtain the total coordination number K . The corresponding failure threshold is $\sigma^{\text{th}} = K\sigma_c X$, where X is a quenched strength variable drawn once per molecule before loading (Eq. 4) and σ_c is the critical stress for bond rupture; we set $\sigma_c = 1$, assuming identical molecules. To simplify the model, both the applied force parameter F and the critical stress σ_c are treated as dimensionless quantities in arbitrary units. Hence, $\sigma_c = 1$ represents the normalized threshold for intermolecular bond rupture, which allows us to focus on relative mechanical behavior rather than absolute physical values.

The strength of each molecule is fixed before loading rather than resampled during it. Molecule i draws a variable $X_i \in [0, 1]$ once, with cumulative distribution

$$P(X_i \leq x) = x^m, \quad \sigma_i^{\text{th}} = K_i \sigma_c X_i, \quad (4)$$

and fails when its mean stress $\sigma_{M,i}$ exceeds its threshold σ_i^{th} . Under this rule the probability that a molecule of coordination K has failed once its mean stress reaches σ is $(\sigma/K\sigma_c)^m$, so Eq. (4) is the original removal probability of our model read as a distribution of strengths; the weakening of a molecule as it loses neighbors, through K , is preserved unchanged. Here m is the Weibull modulus, a dimensionless shape parameter controlling the scatter of the strengths [40, 44]: larger m yields a narrower distribution of rupture strengths, smaller m a broader one. No Weibull modulus has been reported for single collagen fibrils, so we sweep $m \in \{1, 2, 3, 5, 10\}$ and retain $m = 2$, the value used by Parkinson et al. [40], as the illustrative case in the figures. Fig. 5 schematically illustrates the calculation of $\sigma(i)$, σ_M , and K for a representative molecule.

Equations (2)–(4) combine uniform load sharing within each cross-section with local, coordination-dependent molecular resistance. When a molecule is removed, $N(i)$ decreases only in the cross-sections it previously occupied, and the corresponding stress increase is shared uniformly among the remaining segments within each affected cross-section. The

182 removal also reduces K for the molecule's nearest neighbors, lowering their failure thresh-
 183 olds. Together these define a hybrid load-transfer mechanism: stress equilibrates uniformly
 184 among the active segments of each cross-section, while the failure thresholds respond to
 185 local coordination. Two features of the construction set it apart from the idealized limits.
 186 Stress is redistributed without any distance-dependent elastic kernel, so the coordination
 187 channel carries the only local information; and $N(i)$ varies along the fibril while molecules
 188 span different sets of cross-sections, so no single globally shared stress governs the bundle.
 189 The model therefore combines features of equal and local load sharing without mapping
 190 onto either limit.

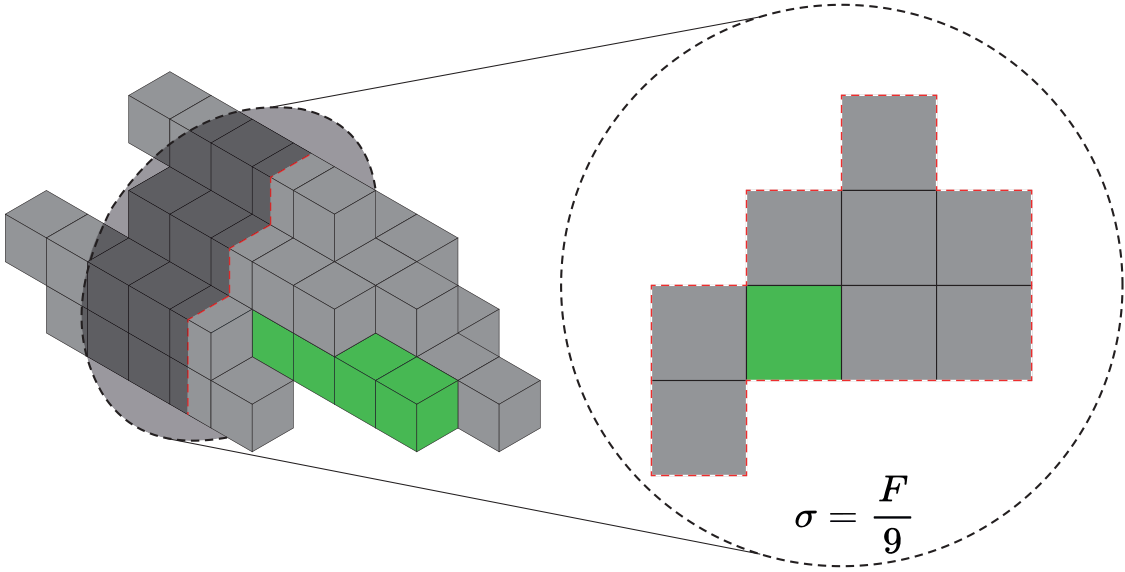


Figure 5. Schematic illustration of the computation of the removal probability P_R for a representa-
 tive molecule (green). Left: a portion of the backbone where each rectangle represents a molecule
 (drawn with five segments for clarity). The green molecule intersects multiple cross-sections; for
 each intersected cross-section i , the local stress is $\sigma(i) = F/N(i)$ (right; example with $N(i) = 9$,
 giving $\sigma(i) = F/9$).

191 Because $\sigma_{M,i} = Fa_i$, with $a_i = \langle 1/N(i) \rangle$ averaged over the cross-sections that molecule i
 192 spans, every molecule has a well-defined failure threshold $F_i^* = K_i \sigma_c X_i / a_i$. The loading is
 193 strictly quasi-static and extremal, ensuring that the entire rupture process proceeds through
 194 a sequence of equilibrium states. The external force is raised continuously until it reaches
 195 the smallest F_i^* present in the current configuration. At this fixed force, the corresponding
 196 molecule fails; stresses and coordinations are then recomputed, and molecules disconnected

from both ends are removed. This iterative update continues at constant force until no remaining molecule exceeds its failure threshold under the updated stress distribution across the fibril. Only after the system fully relaxes into a stable configuration—where every surviving molecule satisfies $F_i^* > F$ —is the external force incremented again. Rupture is defined as the appearance of a completely void cross-section, signaling failure of the backbone. The steps of this process are schematized in Fig. 6.

To capture the statistical nature of this failure process and account for the inherent disorder in molecular strengths, we systematically sampled both the fibril architecture and the strength distribution. Specifically, for each condition of T_s and m , we analyzed an ensemble of 200 distinct fibrils (3×10^4 molecules each) subjected to 50 independent rupture realizations, yielding 10^4 realizations per (T_s, m) combination. Realizations of the same fibril differ solely in the assigned random strengths. Throughout each simulation, we monitored the progress of damage by recording the molecules evicted in every cascade along with the precise external force at which failure occurred.

With this extensive dataset, we quantified the overall mechanical resistance of the fibrils by evaluating the backbone rupture force, F_r , across the different Weibull moduli m . As shown in the main plot of Fig. 7, F_r exhibits a qualitatively similar dependence on T_s ($\log_{10} T_s$) regardless of m . All curves increase monotonically with T_s before reaching a saturation plateau, F_{sat} , whose magnitude grows with m . Fibrils with higher m values are markedly stronger; for instance, at $m = 2$, the mean rupture force increases by an order of magnitude, from 151 ± 36 at $T_s = 2$ to 1532 ± 218 at $T_s = 8192$. Notably, 83% of this total strength gain is already achieved by $T_s = 128$, beyond which further increases in T_s yield negligible enhancements in mechanical resistance. The right inset demonstrates a universal curve collapse when normalizing the rupture force by its saturation value (F_r/F_{sat} versus $\log_{10} T_s$), confirming a scale-invariant behavior independent of m . Finally, the left inset shows that the saturation force F_{sat} grows asymptotically with m , following an exponential dependence accurately fitted by $F_{\text{sat}} = a(1 - e^{-m/b})$.

While these macro-level findings establish that F_r grows systematically with T_s , the global fractal dimension D_f alone—a single static value characterizing the average cross-sectional morphology at assembly—cannot fully capture the underlying failure dynamics. To bridge this global morphological characterization with the local mechanics of rupture, we now track two time-resolved structural quantities: the mean load-bearing cross-sectional count $\langle N \rangle$ and

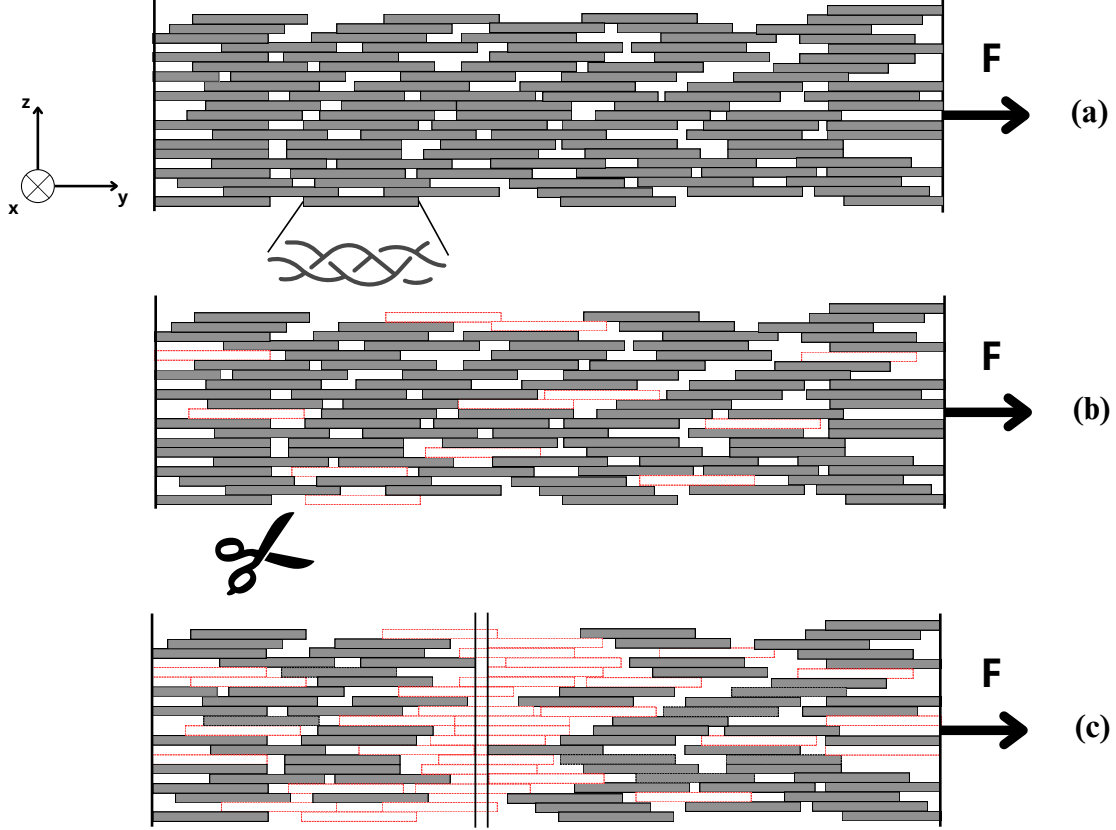


Figure 6. Schematic view of the load-bearing backbone and the stages of the rupture process. (a) Two-dimensional view of the backbone subjected to an axial force F along the y -axis. Each molecule (gray) carries a failure threshold $\sigma^{\text{th}} = K\sigma_c X$ drawn once before loading (Eq. 4); the external force is raised to the smallest failure force in the configuration, and the molecule that reaches it fails. (b) Progressive damage after some molecules have been removed, represented by red dashed lines. (c) Rupture occurs when a cross-section becomes completely empty (solid lines), eliminating the continuous load path.

the mean coordination number $\langle K \rangle$, both normalized by their initial pre-loading values $\langle N_0 \rangle$ and $\langle K_0 \rangle$. Unlike D_f , these are inherently dynamical quantities that evolve as molecules are progressively evicted from the backbone under increasing external force F . Together, they directly govern the local failure threshold of each individual molecule via $F_i^* = K_i \sigma_c X_i / a_i$ (Eq. 4), where $a_i \propto 1/N$. Figure 8 illustrates the dynamic evolution of both quantities across six representative values of T_s at $m = 2$.

The coordination number Fig. 8a decreases monotonically for all architectures. As each molecule fails and is detached, its nearest neighbors lose lateral contacts and their coordi-

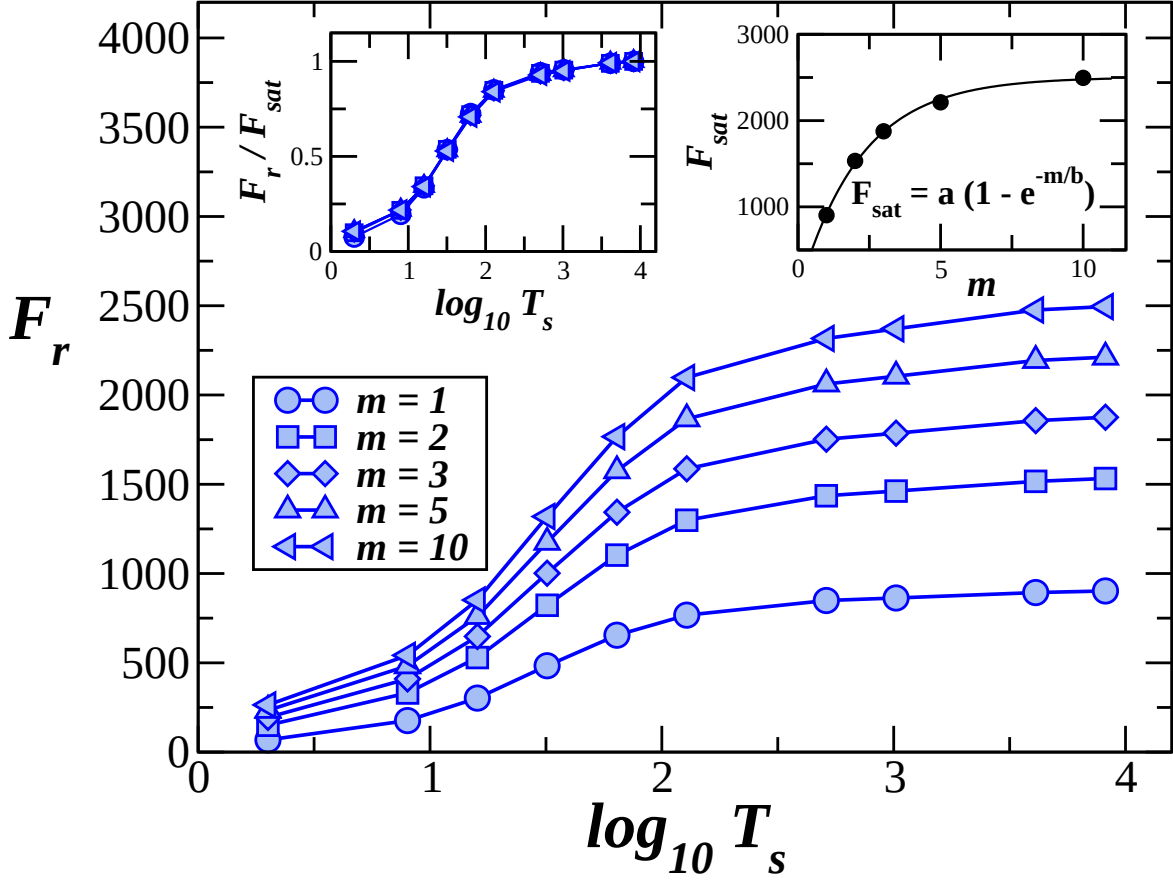


Figure 7. Mean rupture force F_r as a function of T_s for five values of the Weibull modulus m . The right inset shows the curve collapse of F_r/F_{sat} versus T_s , where F_{sat} corresponds to the saturation value of F_r at high T_s values, confirming a universal behavior across different values of m . The left inset shows how F_{sat} behaves as a function of m , where the solid line represents the fit $F_{\text{sat}} = a(1 - e^{-m/b})$ with $a = [\text{value}]$ and $b = [\text{value}]$.

nation K drops accordingly. The force at which each curve terminates identifies the rupture event and grows systematically with T_s , consistent with Fig. 7. More strikingly, the fraction of coordination retained at the moment of rupture is strongly architecture-dependent: $T_s = 2$ loses approximately 20% of $\langle K_0 \rangle$ before failing at $F \approx 200$, whereas $T_s = 128$ retains more than 97% of its initial coordination all the way to $F \approx 1600$. Compact fibrils are therefore not only stronger in absolute terms but also far more resilient at the level of molecular connectivity — their lateral contact network degrades much more slowly per unit of applied force. Since K directly sets the failure threshold via (Eq. 4), this sustained coordination

245 keeps individual thresholds high throughout loading and prevents the runaway weakening
 246 that drives early failure in open architectures.

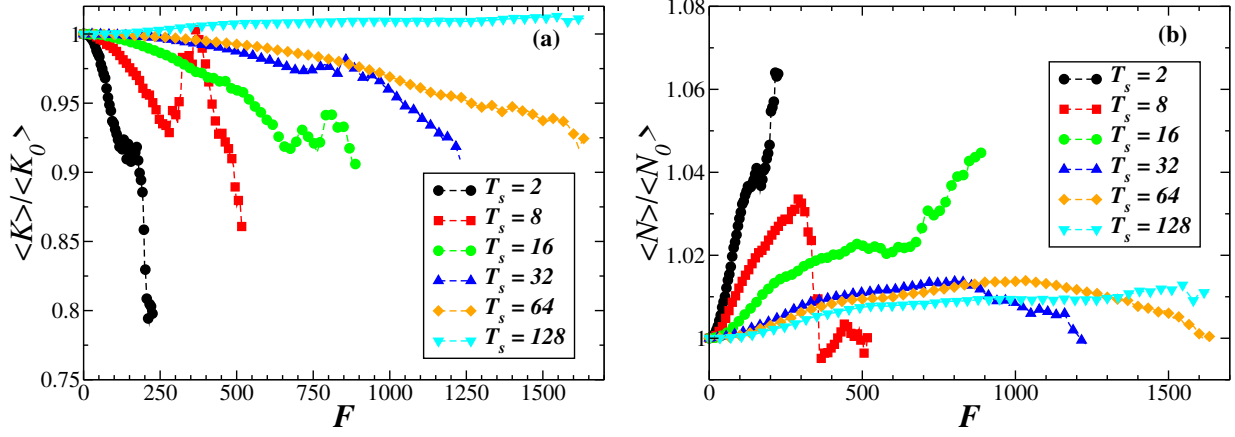


Figure 8. Evolution of the normalized mean coordination number $\langle K \rangle / \langle K_0 \rangle$ (a) and normalized mean load-bearing cross-sectional count $\langle N \rangle / \langle N_0 \rangle$ (b) as functions of the applied force F , for $T_s = 2, 8, 16, 32, 64$, and 128 at Weibull modulus $m = 2$. Quantities are normalized by their respective values immediately before loading, $\langle K_0 \rangle$ and $\langle N_0 \rangle$, and averaged over the ensemble of backbone realizations. Each curve terminates at the mean rupture force F_r of the corresponding architecture.

247 The mean load-bearing area Fig. 8b displays a non-monotonic behavior whose charac-
 248 ter changes markedly with T_s , directly mirroring the morphological transition quantified
 249 by D_f . For open fibrils ($T_s = 2$ and 8), $\langle N \rangle / \langle N_0 \rangle$ rises steeply and irregularly from the
 250 onset of loading, reaching peaks of approximately +6.5% and +3.5% above $\langle N_0 \rangle$, respec-
 251 tively, before falling toward rupture. This initial increase stems from a structural selection
 252 effect rooted in the spatial heterogeneity of open architectures (Figs. 1 and 2): molecules
 253 spanning the thinnest cross-sections $N(i)$ experience disproportionately higher local stresses
 254 $\sigma(i) = F/N(i)$ and fail preferentially at lower forces. Their early removal eliminates these
 255 sparse, high-stress sections from the active backbone, shifting the ensemble average $\langle N \rangle$
 256 upward—acting as an early self-strengthening mechanism where the fibril sheds its weakest
 257 cross-sections to leave behind a denser, more load-capable core. The prominent, irregular
 258 fluctuations along this rise directly reflect the spatial non-uniformity of the fibril itself. As
 259 F continues to grow and this reserve of sparse sections is exhausted, the average turns over
 260 and falls toward rupture. The case $T_s = 16$ marks a transitional regime with a broader peak

261 and attenuated fluctuations, consistent with an intermediate fractal dimension $D_f \approx 1.74$
 262 (Fig. 3). In contrast, for compact fibrils ($T_s \geq 32$), where the fractal dimension D_f shows a
 263 significant shift toward the Euclidean limit ($D_f \rightarrow 2.0$), the cross-sectional packing becomes
 264 nearly uniform. This narrow distribution of $N(i)$ leaves little variance for selective prun-
 265 ing: the large fluctuations disappear, and $\langle N \rangle / \langle N_0 \rangle$ grows smoothly to a mild maximum of
 266 only $\approx +1\%$ before declining gently in the terminal approach to failure. The smoothness of
 267 $\langle N \rangle / \langle N_0 \rangle$ for $T_s \geq 32$ is thus a direct dynamical signature of structural homogeneity, driven
 268 by the same compaction mechanism responsible for the saturation of F_r observed in Fig. 7.
 269 The joint behavior of $\langle K \rangle(F)$ and $\langle N \rangle(F)$ thus provides a coherent microscopic account
 270 of the T_s dependence reported before. Compact fibrils enter the loading protocol with larger
 271 $\langle N_0 \rangle$ and $\langle K_0 \rangle$ — a direct consequence of their higher D_f — and both quantities degrade
 272 more slowly as F increases. Since N controls the local stress and K controls the failure
 273 threshold, the two channels reinforce each other: a larger N lowers $\sigma = F/N$, and a larger
 274 K raises $\sigma^{\text{th}} = K\sigma_c X$. Compact fibrils benefit simultaneously from both effects throughout
 275 the entire loading history, which is the microscopic origin of the order-of-magnitude gain
 276 in rupture force between $T_s = 2$ and $T_s = 8192$. The saturation of F_r visible in Fig. 7
 277 for $T_s \gtrsim 128$ mirrors the saturation of N_0 and K_0 as D_f asymptotically approaches the
 278 Euclidean limit: once the cross-section is nearly space-filling, further increases in T_s yield
 279 diminishing returns in both morphology and mechanical resistance.

280 During rupture, molecules fail in cascades at constant external force. We define the
 281 avalanche as the complete cascade released by one increase of the force: the molecules
 282 whose thresholds are exceeded, together with those that lose their continuous load path as
 283 a consequence, all removed before the force is raised again. This is the set of molecules that
 284 fail between two consecutive increases of the external force, and it is the standard avalanche
 285 of fiber-bundle models, a form of crackling dynamics in driven disordered systems [45–50].

286 We analyze this collective failure process through the framework of avalanche statistics in
 287 driven disordered systems [26, 27, 51]. Here, the avalanche size s corresponds to the number
 288 of molecules evicted in a single cascade, as defined above. Figure 9 displays the pre-terminal
 289 cascade size distribution $P(s)$ on logarithmic axes for $T_s = 2, 8, 16$, and 128 at $m = 2$. Across
 290 all conditions, $P(s)$ exhibits a power-law regime over intermediate cascade sizes, followed by
 291 a rapid, stretched-exponential cutoff at large s . Similar scaling behaviors followed by sharp
 292 cutoffs have been documented in numerical studies of impact fragmentation and structural

breakdown [52–55]. Based on these features, $P(s)$ obeys the scaling ansatz:

$$P(s) \propto s^{-\gamma} \exp \left[- \left(\frac{s}{s_c} \right)^\eta \right] \quad (5)$$

where γ is the power-law scaling exponent, while s_c and η represent the characteristic cutoff size and stretching exponent, respectively. The solid lines in Fig. 9 show the best non-linear fits of Eq. (5) to the data across both the scaling and cutoff regimes for $T_s = 2$ ($\gamma = 2.00 \pm 0.01$, $s_c = 124 \pm 5.67$, $\eta = 3.53 \pm 0.26$) and $T_s = 128$ ($\gamma = 2.55 \pm 0.008$, $s_c = 67 \pm 1.65$, $\eta = 2.71 \pm 0.21$).

A closer examination of $P(s)$ reveals two distinct structural regimes as a function of T_s . At small avalanche sizes the distributions for all values of T_s are nearly indistinguishable, indicating that single- and few-molecule failure events are governed purely by localized threshold dynamics and remain insensitive to the global fibril architecture. In the intermediate-to-large size regime, however, the curves diverge into power-law regimes which depends on the T_s . For the most open architecture ($T_s = 2$), the distribution follows a shallower decay with $\gamma = 2.00 \pm 0.013$, signaling a significantly higher probability of large pre-terminal cascades. Conversely, for more compact architectures ($T_s \geq 128$), the distributions collapse onto a single, steeper power law with $\gamma = 2.55 \pm 0.008$, reflecting a systematic suppression of large-scale failure events. Notably, this scaling crossover occurs abruptly between $T_s = 2$ and $T_s = 128$, directly mirroring the rapid initial rise of the fractal dimension D_f away from the DLA limit (Fig. 3). For values above $T_s = 128$ the exponent γ keep approximately constant within the error bars (data not show) which are in agreement with results for D_f , $\langle N \rangle$ and $\langle K \rangle$ that reach a saturation values for $T_s \geq 128$.

This dynamical transition has a clear physical origin rooted in the local structural properties discussed in the previous section. In open fibrils ($T_s = 2$, $D_f \approx 1.71$), the high spatial heterogeneity gives rise to a broad distribution of cross-sectional counts $N(i)$, as evidenced by the large fluctuations in $\langle N \rangle(F)$ (Fig.8b). Molecules in the sparsest cross-sections experience severe local stress concentrations $\sigma(i) = F/N(i)$; upon their failure, the redistributed load readily overloads equally sparse neighboring sections, facilitating cascade propagation. Compounding this effect, the rapid loss of coordination $\langle K \rangle$ observed at $T_s = 2$ —a reduction of nearly 20% prior to rupture (Fig.8a)—progressively degrades the failure thresholds $F_i^* = K_i \sigma_c X_i / a_i$ of surviving molecules, making them increasingly prone to sequential collapse. Together, the high variance in $N(i)$ and the rapid decay of K sustain extensive cas-

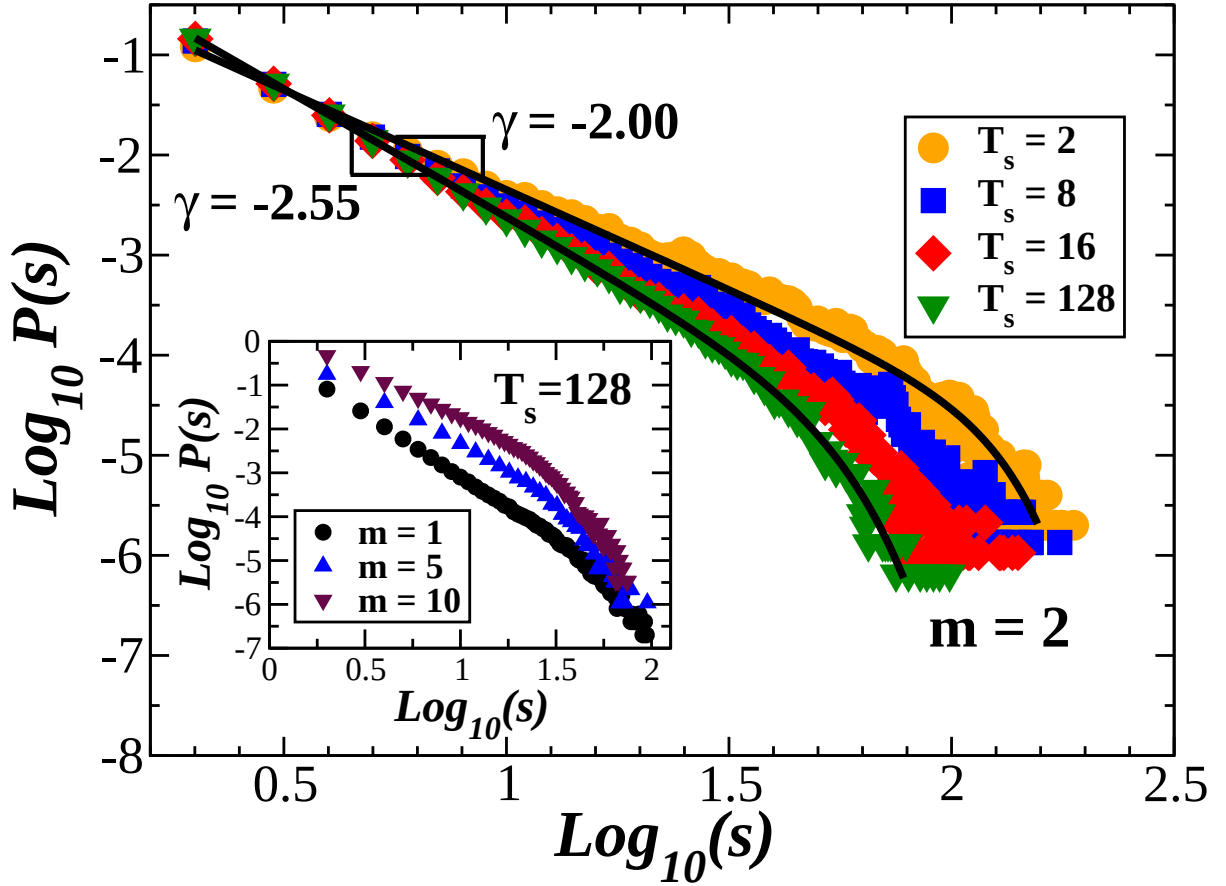


Figure 9. Avalanche size distribution $P(s)$ on logarithmic axes for $T_s = 2, 8, 16$, and 128 at Weibull modulus $m = 2$. All curves display a power-law regime followed by a stretched-exponential cutoff. Solid lines are fits of the form $P(s) \propto s^{-\gamma} \exp\left[-\left(\frac{s}{s_c}\right)^\eta\right]$ performed for the cases $T_s = 2$ ($\gamma = 2.00 \pm 0.013$, $\eta = 3.53$, $s_c = 124.7$) and $T_s = 128$ ($\gamma = 2.55 \pm 0.008$, $\eta = 2.71$, $s_c = 51.61$). Inset: $P(s)$ for $T_s = 128$ and $m = 1, 5$, and 10 . Curves in the inset have been shifted vertically for visual clarity.

324 cades, resulting in the shallower exponent $\gamma = 2.00$ and extending the stretched-exponential
325 cutoff up to $s \approx 200$.

326 For compact fibrils ($T_s \geq 32$), both mechanisms are strongly suppressed. The uniform
327 cross-sectional packing associated with $D_f \rightarrow 2.0$ leads to a smooth, low-amplitude evo-
328 lution of $\langle N \rangle(F)$ (Fig. 8b) and maintains the coordination number near its initial value
329 $\langle K_0 \rangle$ throughout loading (Fig. 8a). This structural homogeneity ensures an even stress dis-
330 tribution and preserves molecular failure thresholds, effectively arresting cascade growth.

331 Consequently, the power-law exponent steepens to $\gamma = 2.55$ and the cutoff shifts to much
 332 smaller scales ($s \approx 30\text{--}50$). This sharp reduction confirms that the maximum pre-terminal
 333 cascade size is ultimately dictated by the local structural organization of the backbone rather
 334 than the total size of the system.

335 Although our coarse-grained framework adopts necessary physical simplifications—such
 336 as representing collagen molecules as rigid rods with a reduced aspect ratio of 18:1 [21, 40],
 337 omitting rotational diffusion, and capturing complex biochemical regulation through an ef-
 338 fective control parameter T_s [11, 29, 56–59]—these approximations are essential to make the
 339 multi-scale physics computationally tractable while retaining its core mechanics. Rather
 340 than aiming for exact quantitative metrics or reproducing detailed bond-breaking mechan-
 341 ics [60–62], the primary strength and novelty of this model lie in its ability to isolate the
 342 fundamental, qualitative relationships between stochastic assembly topology and mechanical
 343 resilience. Within this transparent framework, we demonstrate that structural compaction
 344 directly strengthens the fibril backbone by modulating the local load-bearing area $N(i)$
 345 and preserving the molecular coordination number K , showing that mechanical resilience is
 346 achieved without requiring a complex, self-similar load-transfer mechanism.

347 Importantly, despite these simplifying assumptions, the model’s core dynamical predic-
 348 tion—that rupture proceeds via avalanche-like, pre-terminal failure cascades—is strongly
 349 supported by experimental evidence. Because avalanche propagation occurs faster than im-
 350 posed loading rates [39], single-molecule force-spectroscopy and tensile tests on individual
 351 collagen microfibrils consistently observe discrete force and stress drops resulting from in-
 352 termolecular cross-link ruptures [24, 63, 64], directly validating the physical relevance of
 353 our cascade framework. By successfully bridging the gap between stochastic fractal as-
 354 sembly and local damage mechanics, this work provides a robust, mechanistic baseline for
 355 understanding how geometric packing and spatial heterogeneity govern collective failure in
 356 biological tissues, offering valuable insights for both biomechanics and the rational design of
 357 resilient, bio-inspired synthetic materials.

358 **III. CONCLUSION**

359 In this work, we coupled a three-dimensional diffusion-limited aggregation model of col-
 360 lagen fibrillogenesis with a progressive probabilistic rupture framework to establish a mech-

anistic link between assembly kinetics, fibril architecture, and mechanical failure. The post-attachment surface diffusion parameter T_s acts as a single control variable that simultaneously governs morphological organization and fracture resistance. As T_s increases, the fractal dimension D_f rises monotonically from $D_f = 1.708 \pm 0.005$ ($T_s = 2$) to $D_f = 1.963 \pm 0.001$ ($T_s = 8192$), bridging open fractal assemblies to nearly compact, space-filling architectures. Crucially, D_f sets the pre-loading structural foundation: higher D_f directly yields a larger initial load-bearing area $\langle N_0 \rangle$ and mean coordination number $\langle K_0 \rangle$, granting compact fibrils an immediate mechanical advantage.

During the rupture process, the interplay among D_f , $\langle N \rangle$, and $\langle K \rangle$ unveils the microscopic origin of the order-of-magnitude strength gain observed between open and compact architectures. In open fibrils ($T_s = 2, D_f \approx 1.71$), severe stress concentration at sparse cross-sections induces early localized failures. While pruning these thinnest sections produces a transient 6.5% self-strengthening rise in $\langle N \rangle / \langle N_0 \rangle$, the rapid $\sim 20\%$ depletion of coordination $\langle K \rangle$ degrades molecular thresholds and sustains extensive cascade propagation ($\gamma = 2.00 \pm 0.013, s_c \approx 124$). Conversely, compact fibrils ($T_s \geq 128, D_f \rightarrow 2.0$) feature uniform cross-sectional packing that mitigates stress concentrations, restricts the self-strengthening rise to $\approx 1\%$, and preserves $> 97\%$ of $\langle K_0 \rangle$ up to failure. This structural homogeneity suppresses large pre-terminal cascades, steepening the power-law exponent to $\gamma = 2.55 \pm 0.008$ and shifting the cutoff to $s_c \approx 50$.

From a biological and engineering perspective, D_f serves as an architectural bridge linking assembly conditions to failure statistics, demonstrating how biological systems can tune tensile strength through post-attachment relaxation—with 83% of the total strength gain already realized at $T_s = 128$. While physical simplifications (rigid rods, reduced aspect ratio, and effective control parameters) delimit the present framework, they are essential to isolate core geometric and topological principles from the vast parameter space of fibrillogenesis. By establishing clear scaling relations and falsifiable predictions, this work provides a mechanistic theoretical baseline against which single-fibril force-spectroscopy measurements can be calibrated, advancing the rational design of mechanically optimized bio-inspired fibers.

IV. AUTHOR CONTRIBUTIONS

All authors contributed equally to this work.

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